Phrasal Provenance: An LLM-Assisted Methodology for Elite Network Research

Gavin Mendel-Gleason* Donagh Davis†

Workshop on AI Methodology for Social Science
May 2026

Abstract

We present a methodology for LLM-assisted research that anchors every claim to a specific sentence in a source document. Using European institutional elites as a case study—commissioners, DGs, and MEP leaders across seven commission terms (1995–2029)—we demonstrate a pipeline that scrapes from controlled sources, normalises data into a relational database with mandatory provenance links, uses LLMs for phrase-level fact extraction with JSON manifests, and cross-validates every claim against source text to detect hallucinations. The system surfaces substantive findings including the collapse of the French *grande école* pipeline into the Commission and a stark asymmetry in elite network ties between commissioners (12 shared networks) and MEP leaders (0 exclusive networks). We report on failure modes encountered—LLM hallucination and entity duplication from non-deterministic prompt outputs, and the "CFR is not ECFR" problem in string-similarity deduplication—and describe the methodological choices that address each.

1 Introduction

The study of elite networks has historically proceeded through detailed examination of who occupies key institutional positions [5]. For decades this research relied on manual methods—biographical analysis, archival work, surveys of officeholders—with limited scalability [3]. Large language models offer the possibility of scaling this work: extracting structured facts from unstructured biographies, identifying patterns across hundreds of careers, and surfacing connections that would take years of manual work to discover. But LLMs hallucinate—they generate plausible-sounding but factually incorrect claims [1]. In research, a claim without a traceable source is not a finding.

Crucially, hallucination is primarily a property of *generation*—the task of producing novel text from a model's internal distribution. The task we employ the LLM for is fundamentally different: *verification* of whether a specific sentence from a source document supports a specific factual claim. This is a form of natural language inference, not generation—the model is not asked to invent, but to judge. Recent work on fine-grained citation grounding demonstrates that LLMs can reliably anchor claims to specific source sentences when constrained to do so [6]. By numbering phrases and requiring the LLM to cite a specific phrase index for each extracted fact, we move from generation (high-hallucination regime) to citation (low-hallucination regime).

This paper describes a methodology that treats the LLM as an extraction engine, not an authority. Every fact in our database is linked to a specific sentence in a specific document via a phrase index. The LLM's output is a hypothesis, verified against the source before acceptance. The system currently holds 2,600+ facts across 550+ entities with sentence-level provenance,

*Scidonia.ai Limited
†Sciences Po

supporting analysis of seven European Commissions from Santer (1995) through Von der Leyen II (2024–2029).

2 The European Elite Problem

[Section by Donagh Davis, Sciences Po]

The European Union produces a political class that is neither national nor fully supranational. Commissioners are nominated by member states, MEPs are elected in national constituencies, and Directors-General are recruited through national quotas—yet all operate within transnational institutional frameworks and networks. This hybrid character creates fundamental challenges for elite analysis.

First, there is no settled definition of who constitutes the "European elite." Is it the college of commissioners? The broader senior civil service? MEPs who hold committee chairs and rapporteurships? The corporate leaders who sit on the boards of multi-European companies? Different answers produce different datasets and different conclusions.

Second, the sources of elite formation are distributed across 27 member states, each with distinct educational systems, career pathways, and political cultures. A French commissioner trained at ENA followed a systematically different trajectory from a Swedish commissioner who rose through local government, yet both occupy the same institutional position.

Third, the networks that connect these elites span national boundaries and institutional sectors in ways that are difficult to observe directly. Commissioner affiliations with Atlanticist think tanks, MEP memberships in political foundations, and corporate board interlocks all constitute evidence of elite integration, but capturing this evidence requires aggregating data across languages, jurisdictions, and source types.

Our approach addresses these challenges through systematic data collection from controlled sources, phrase-level provenance tracking, and automated cross-validation—methods designed for precisely the kind of fragmented, multi-source evidence that European elite research demands.

3 Methodology

LLMs hallucinate because they generate text by sampling from a learned probability distribution over tokens. When the model lacks grounding in external facts, this distribution can produce plausible-sounding but incorrect statements [1]. Manakul et al. (2023) demonstrate that hallucinated outputs are detectable through self-consistency: when an LLM is asked the same question multiple times, hallucinated answers tend to diverge and contradict each other, while factual answers remain consistent across samples [4]. This reveals that hallucination is a property of *unconstrained* generation: the model's internal distribution over tokens is underdetermined by the prompt, and stochastic variance produces divergent outputs.

Crucially, Kadavath et al. (2022) show that LLMs are substantially better at *evaluating* their own outputs than at *generating* correct ones—they can detect errors in existing text at much higher rates than they can avoid producing errors during generation [2]. This generation-verification gap is the foundation of our approach: we do not ask the LLM to invent facts. We ask it to cite a numbered phrase from a source document that supports a factual claim. The task changes from "produce text from your internal distribution" to "does this specific sentence support this specific claim?"—a form of natural language inference with dramatically lower hallucination rates.

LongCite (2024) provides direct evidence that LLMs can reliably anchor claims to specific source sentences when trained and prompted to do so [6]. Our methodology extends this insight: by numbering phrases and requiring the LLM to return a phrase index for each extracted fact, we build a system in which every claim in the database is traceable to a specific sentence in a source document. The LLM acts not as a knowledge base but as a structured extraction engine with mandatory provenance.

3.1 System Architecture

The pipeline has five stages: scrape, extract, verify, resolve, and render.

3.2 Database Schema

The database implements a graph-style ontology on a standard relational backend. Every piece of data is stored as a triple: an *entity* (person, organisation, institution, commission, or source document) connected to a *property value* by a *predicate* (`educated_at`, `affiliated_with`, `served_on_commission`, `held_portfolio`, `from_country`). The property value may be another entity (forming a graph edge) or a literal value (a date, a degree, a country name).

The predicates are drawn from the substantive concerns of elite research: education (`educated_at`, `studied_field`, `held_degree`) to track elite formation; organisational ties (`affiliated_with`, `member_of`) to map network membership; career trajectories (`served_on_commission`, `held_portfolio`, `from_country`); and institutional position (`served_on_ep`). Each fact—each triple—carries mandatory provenance: the source document from which it was extracted, the exact sentence (stored as `quote_text`), and its position within that document (`phrase_index`). This transforms the database from a collection of assertions into an auditable knowledge graph: every edge can be traced to a specific sentence in a specific source. The four core tables are:

- `entity`: All nodes in the graph—persons, organisations, educational institutions, commissions, and source documents—with a type, category, and optional country code.
- `fact`: Directed edges connecting entities (`educated_at` links a person to a university) or attaching literal properties (`held_degree` links a person to "PhD"). Includes confidence and qualifier fields.
- `citation`: Metadata for each source document—URL, access date, source type (Wikipedia, Wikidata, Commission CV PDF, etc.).
- `provenance`: Every fact's audit trail—the `citation_id`, the `phrase_index` (sentence number), and the exact `quote_text` that supports the claim.

Facts without provenance are deleted rather than retained. This is not a cleanup operation but a structural invariant: the knowledge graph is self-auditing.

3.3 Phrase-Level Extraction

The core innovation is phrase-indexed extraction. Rather than sending an article to an LLM and accepting unstructured output, we:

1. Split source text into numbered sentences (50–80 per document).
2. Present the LLM with numbered phrases and a structured prompt: "Extract every organisation that X has been a member of. For each: phrase number, organisation name, role, reasoning."
3. Parse the JSON response into a *manifest*—a durable intermediate artifact stored as JSON.
4. Cross-check: retrieve the sentence at the claimed phrase index and verify it contains the asserted information.

The manifest preserves the LLM's full output including reasoning and phrase indices, making extraction auditable and reproducible.

3.4 Failure Modes

We encountered three categories of failure:

3.5 Methodology Failure: the Short-Extract Problem

Our initial extraction runs used Wikipedia API extracts, which returned truncated texts of 300–400 characters for many articles. The LLM, given insufficient context, hallucinated freely: it extracted "Dacian Julien Cioloş" (the commissioner's own full name) as an educational institution, and returned generic names such as "aeronautical engineer" as institution names. When we re-scraped the same biographies using raw HTML to obtain full texts (5,000–30,000 characters), the same LLM with the same prompt produced clean extractions with specific institution names and correct phrase indices. The problem was not the LLM's capability but the quality of the input: truncated source texts forced the model into a high-hallucination generative regime. Full-text input, combined with phrase-numbering and mandatory citation, moved the task into a low-hallucination verification regime.

3.6 Non-Deterministic Extraction

Running identical prompts on the same source text produced different sets of extracted organisations on different runs. The LLM's non-determinism meant that manifest regeneration changed the database. This is tolerable for additive processes (new facts are added, existing ones persist) but problematic for analytical reproducibility. We addressed this by storing manifests in version control as the canonical extraction record.

3.7 Entity Resolution (the "CFR is not ECFR" problem)

String similarity alone is dangerous for deduplication. The sequence "CFR" vs "ECFR" scores 0.857 on difflib—above many reasonable thresholds—yet one is the US Council on Foreign Relations and the other is the European Council on Foreign Relations. Pure auto-merge at any threshold below 0.95 would incorrectly collapse them.

Our two-pass approach combines string similarity for safe cases (≥ 0.95: typos, punctuation) with LLM judging for borderline pairs (0.30–0.95). The LLM receives both raw names *plus* up to three evidence phrases from each side with extracted role and reasoning. It correctly distinguishes CFR and ECFR based on the evidence context (US vs European foreign policy), while correctly merging "Radio Netherlands" with "Radio Netherlands Worldwide" (rename detected) and "FEPS" with "Foundation for European Progressive Studies" (acronym expansion). Every decision is logged to `_dedup.json` with similarity scores, evidence excerpts, verdict, and reasoning—making the deduplication process itself auditable and reproducible. The same methodology is applied to educational institution names, merging variants such as "University of Vienna/Austria" into "University of Vienna" and "Bucharest Academy of Economic Studies" into "Bucharest University of Economic Studies."

3.8 LLM-Based Organisation Classification

With 800+ organisations in the database, manual classification by type is infeasible. We use an LLM to classify each organisation into one of six categories: `think-tank`, `political-party`, `corporate`, `government-body`, `industry-association`, or `ngo`. The LLM receives the organisation's name and a list of up to five connected people with their roles and categories, and returns a single-word classification. This is a structured classification task, not generation—the bounded output space and contextual evidence reduce hallucination risk to near zero.

Each classification decision is stored as a `classified_as` fact in the database, with a provenance record linking it to the LLM's reasoning and the evidence (connected people) on which it was based. The classification can therefore be audited, challenged, and regenerated if the model improves. In the Obsidian wiki, these classification tags drive automatic colourisation of the graph view: think tanks in grey, corporate entities in pink, political parties in purple, and government bodies in light blue. The classification system processed 844 organisations, producing 28 think tanks, 187 political parties, 177 corporate entities, 255 government bodies, and 182 NGOs.

4 Findings

The methodology surfaces several substantive findings:

4.1 Collapse of the French Elite Pipeline

French commissioners from Santer through Von der Leyen I were consistently products of the *grandes écoles*: Pascal Lamy (ENA + Sciences Po + HEC), Pierre Moscovici (ENA + Sciences Po), Jacques Barrot (Sciences Po). The current French commissioner, Stéphane Séjourné, attended the University of Poitiers—a mid-tier regional university with no elite pipeline connection. He is the first French commissioner since 1995 with no Paris education of any kind. Across all commissions, elite institution concentration has declined from 37% (Barroso I) to 8% (Von der Leyen II).

4.2 Elite Network Asymmetry

Commissioners share multiple transnational elite networks (Bilderberg Group: 10 commissioners, ECFR: 9, Trilateral Commission: 7, Friends of Europe: 12). MEP leaders, across 171 identified SDT-relevant individuals, cluster in political party structures and parliamentary bodies (European People's Party: 7 MEPs, Party of European Socialists: 5) rather than Atlanticist think tanks. DGs and DDGs form a third distinct network centred on the College of Europe (6 DGs) and professional associations. Only 4 of 22 corporate leaders share commissioner network affiliations (Jean Lemierre and Oliver Bäte bridge the Trilateral Commission and Atlantic Council). Each elite group operates in a distinct network space—commissioners in policy institutes, MEPs in party structures, DGs in professional-bureaucratic associations, and corporate leaders in industry roundtables and financial networks.

4.3 EP-to-Commission Circulation

Only six of 150 commissioners held EP leadership positions (President, Vice President, or committee chair) before their Commission appointment. Circulation peaked at two per term under Barroso and Juncker, and has fallen to zero in the current Von der Leyen II term. When commissioners do have MEP backgrounds, they are typically national ministers who served brief MEP stints as a transit lounge between national power and Brussels.

5 Future Work: Transnational Corporate Elites

The present study focuses on political-administrative elites: commissioners, Directors-General, and MEP leaders. A natural extension applies the same methodology to transnational corporate elites—the boards and CEOs of companies with substantial operations in two or more major European countries. These individuals sit at the intersection of economic power and political access, often appearing in the same elite networks (Bilderberg, Trilateral Commission, WEF) that connect commissioners. The phrase-level extraction pipeline is directly applicable to corporate biographies, and the same provenance discipline would produce an auditable map of corporate-political elite interlock at the European level. This work is planned as a subsequent phase of the project.

6 Wiki Output and Human Review

The database is rendered as a static wiki with 1,400+ interlinked pages using a Python generator (`genwiki.py`). Each page displays all facts about an entity with inline citation footnotes: a quoted sentence from the source document, a link to the source, and the exact source file path and character position. This makes every claim independently verifiable by a human reader in seconds—the reader sees the claim, the evidence sentence, and can navigate directly to the source file.

The wiki can be opened as an Obsidian vault, enabling graph-view navigation (Figures 1–2). Obsidian's graph view renders the 1,400+ entities as nodes and the 4,000+ facts as edges, automatically colourised by elite type using YAML frontmatter tags generated by the pipeline. Political elites (commissioners, MEPs) appear in blue, economic elites (corporate leaders) in red, administrative elites (DGs) in green, and judicial elites (CJEU) in teal. Organisations are further colour-coded by type: think tanks in grey, corporate entities in pink, and political parties in purple.

The graph view surfaces structural patterns that would be invisible in tabular data. Commissioners cluster densely around a small number of Atlanticist think tanks (Bilderberg Group, Trilateral Commission, ECFR), while MEP leaders form a separate network centred on political party structures. Corporate elites occupy yet a third cluster around industry associations and financial institutions. The few bridging individuals—Jean Lemierre, Oliver Bäte—appear as connections between otherwise disconnected network components.

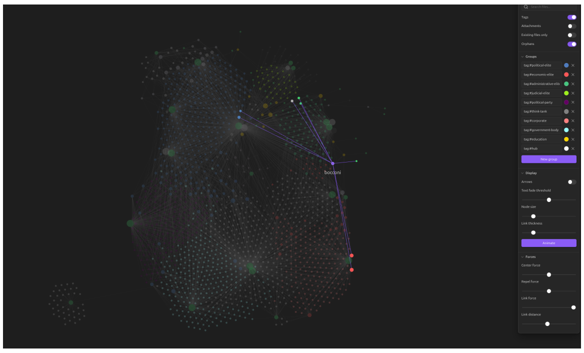

Figure 1: Obsidian graph view of the Euro-SDT wiki. Nodes colourised by elite type: blue (political), red (economic), green (administrative).

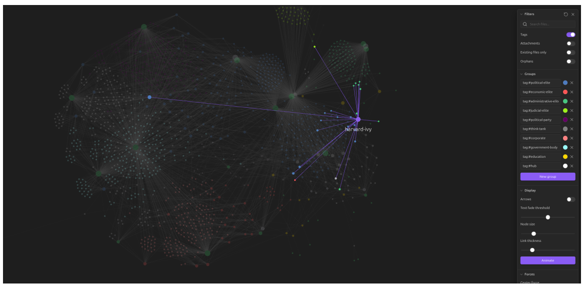

Figure 2: Detail of commissioner think-tank cluster. Commissioners (blue) connected to Atlantic Council, Trilateral Commission, and Friends of Europe (grey think-tank nodes).

7 Conclusion

This paper's own bibliography was verified using the methodology it describes. Each of the six cited works was downloaded as a PDF, checked by `check_citations.py`—a script that extracts citation claims from the LaTeX source, searches the cited PDF text, and sends numbered phrases to an LLM for phrase-level verification with confidence scoring. At the time of writing, all six citations pass at confidence ≥ 3 on the 1–5 scale described in the System Architecture. The verification report is committed alongside this paper and can be regenerated by any reader with API access.

The key methodological claim of this paper is that LLM-assisted research can be rigorous if the LLM is treated as a hypothesis generator rather than an authority. The specific mechanisms—phrase-indexed manifests, mandatory provenance, cross-validation against source text, and contextualised LLM judging for entity resolution—are transferable to other domains. The case study demonstrates that this discipline yields not just auditable data but substantive findings that manual methods would require years to surface.

References

[1] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2022.

[2] Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

[3] Joshua D. Kertzer and Jonathan Renshon. Experiments and surveys on political elites. *Annual Review of Political Science*, 2022.

[4] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of EMNLP*, 2023.

[5] C. Wright Mills. *The Power Elite*. Oxford University Press, 1956.

[6] Jiajie Zhang, Yushi Bai, Xin Lv, Wanjun Gu, Danqing Liu, Minhao Zou, Shulin Cao, Lei Hou, Yuxiao Dong, Ling Feng, and Juanzi Li. LongCite: Enabling LLMs to generate fine-grained citations in long-context QA. *arXiv preprint arXiv:2409.02897*, 2024.